

Immunology

Ingyu Bahng

June 2026

Contents

1 Immune System Players	3
2 Immune System Organs	3
3 Innate Immunity Processes	3
3.1 Inflammation	3
3.2 Innate Pathogen Pattern Recognition	4
3.3 NOX Complexes	4
4 Antibodies	4
4.1 Immunoglobulin Structure & Isotypes	4
4.2 B Cell Receptors	4
4.3 Immunoglobulin; DNA to Expression/Secretion	4
4.4 Immunoglobulin Genetics	4
4.5 Other	5
5 B Cells	5
5.1 Naive B Cell Development	5
5.2 B Cell Activation and Proliferation	5
5.2.1 Thymus Independent and Dependent B Cell Activation	5
5.2.2 Antibody Development	5
5.2.3 Plasma vs Memory B Cells	6
5.3 Leukemia & Lymphoma	6
6 Complement System	6
6.1 MAC Complex Attack	6
6.2 Complement Pathways	6
7 Major Histocompatibility Complex	7
7.1 MHC Structure & Genetics	7
7.2 MHC Antigen Preparation	7
8 T Cells	7
8.1 T Cell Receptor	7
8.1.1 TCR Complex Characteristics	7
8.1.2 CD4 & CD8	8
8.1.3 TCR Genes and Rearrangement	8
8.2 T Cell Development	9
8.3 TCR Antigen Response	9
9 Cytokines and Signals	9

9.1	Receptor Signaling Types	9
9.2	Immune Signaling Types & Themes	10
9.3	Cytokine Interaction Types	10
10	Inflammation	10
11	Tolerance	10
12	Autoimmunity	10
13	Hypersensitivity	10
14	Transplant Immunity	10

1 Immune System Players

Core Concept 1.1: Hematopoietic Stem Cells (HSCs) & Lineage

Core Concept 1.2: Lymphoid Lineage

Core Concept 1.3: Myeloid Lineage

Core Concept 1.4: Follicular Dendritic Cells

2 Immune System Organs

Core Concept 2.1: Primary Immune System Organs

Core Concept 2.2: Secondary Immune System Organs

Core Concept 2.3: Tertiary Immune System Organs

3 Innate Immunity Processes

3.1 Inflammation

Core Concept 3.1: Fever & The Local Response

Core Concept 3.2: Immune Cell Extravasation

3.2 Innate Pathogen Pattern Recognition

Core Concept 3.3: Pathogen Associated Molecular Patterns (PAMPs)

Core Concept 3.4: Pattern Recognition Receptors (PPRs)

3.3 NOX Complexes

Core Concept 3.5: NADPH Oxidase (NOX) Complexes

4 Antibodies

4.1 Immunoglobulin Structure & Isotypes

Core Concept 4.1: Immunoglobulin Structure

Core Concept 4.2: Immunoglobulin Isotypes (<i>H Chain Differences</i>)

4.2 B Cell Receptors

Core Concept 4.3: B Cell Receptors

4.3 Immunoglobulin: DNA to Expression/Secretion

Core Concept 4.4: Immunoglobulin: DNA to Expression/Secretion

4.4 Immunoglobulin Genetics

Core Concept 4.5: L Chain Gene Sequences

Core Concept 4.6: H Chain Gene Sequences

Core Concept 4.7: L/H Chain Gene Expression**Core Concept 4.8: RSS Joining Mechanism****4.5 Other****Core Concept 4.9: History: *Edelman & Porter*****Core Concept 4.10: Haptens (*Small Antigens*)****Core Concept 4.11: Monoclonal Antibodies (mAb)****5 B Cells****5.1 Naive B Cell Development****Core Concept 5.1: Naive B Cell Development****5.2 B Cell Activation and Proliferation****5.2.1 Thymus Independent and Dependent B Cell Activation****Core Concept 5.2: Thymus Independent B Cell Activation****Core Concept 5.3: Thymus Dependent B Cell Activation****5.2.2 Antibody Development****Core Concept 5.4: Lymph Node Follicle Development**

Core Concept 5.5: Affinity Maturation**Core Concept 5.6: Class Switching****5.2.3 Plasma vs Memory B Cells****Core Concept 5.7: Plasma Cells****Core Concept 5.8: Memory Cells****5.3 Leukemia & Lymphoma****Core Concept 5.9: Leukemia****Core Concept 5.10: Lymphoma****6 Complement System****6.1 MAC Complex Attack****Core Concept 6.1: MAC Complex Attack****6.2 Complement Pathways****Core Concept 6.2: Classical Pathway****Core Concept 6.3: Lectin Pathway****Core Concept 6.4: Alternative Pathway**

Core Concept 6.5: Side Signal Pathways & Other

7 Major Histocompatibility Complex

7.1 MHC Structure & Genetics

Core Concept 7.1: MHC Characteristics**Core Concept 7.2: MHC Gene Sequences****Core Concept 7.3: MHC Mendelian Genetics**

7.2 MHC Antigen Preparation

Core Concept 7.4: Endogenous MHC I (T_c Cells)**Core Concept 7.5: Exogenous MHC I / Cross Presentation (T_c Cells)****Core Concept 7.6: Exogenous MHC II****Core Concept 7.7: Exogenous CD1 MHC**

8 T Cells

8.1 T Cell Receptor

8.1.1 TCR Complex Characteristics

Core Concept 8.1: $\alpha\beta$ and $\gamma\delta$ Receptors

Core Concept 8.2: $\gamma\epsilon$ and $\delta\epsilon$ Heterodimers**Core Concept 8.3: $\zeta\zeta$ or $\zeta\eta$ Heterodimers****8.1.2 CD4 & CD8****Core Concept 8.4: CD4****Core Concept 8.5: CD8****8.1.3 TCR Genes and Rearrangement****Core Concept 8.6: Gene Sequence and Receptor Overview****Core Concept 8.7: α Rearrangement****Core Concept 8.8: β Rearrangement****Core Concept 8.9: γ Rearrangement****Core Concept 8.10: δ Rearrangement****Core Concept 8.11: Subsequent Expression****Core Concept 8.12: History: *Hendrick and Davis***

8.2 T Cell Development

Core Concept 8.13: Developmental Pathway

8.3 TCR Antigen Response

Core Concept 8.14: Antigen Response Pathway

Core Concept 8.15: TCR Antigen Recognition to ZAP70

Core Concept 8.16: Lipid Rafts

9 Cytokines and Signals

9.1 Receptor Signaling Types

Core Concept 9.1: Immunoglobulin

Core Concept 9.2: Type I Receptors

Core Concept 9.3: Type II Receptors

Core Concept 9.4: Interleukin 17

Core Concept 9.5: Tumor Necrosis Factor

Core Concept 9.6: Chemokines

9.2 Immune Signaling Types & Themes

Core Concept 9.7: Immune Signaling Types

Core Concept 9.8: Signaling Themes

9.3 Cytokine Interaction Types

Core Concept 9.9: Cytokine Interaction Types

10 Inflammation

11 Tolerance

12 Autoimmunity

13 Hypersensitivity

14 Transplant Immunity

Rice University - Immunology - U1 - Innate Immunity and B Cell Function - U2 - Complement, MHC I and II, T Cells, and Cytokines - U3 - Inflammation, Tolerance, Autoimmunity - U4 - Dueling with Pathogens and Cancer